Imitation Learning

Kianté Brantley
CS 1840 Fall 2026

Announcements

HW3 was due last Friday!

HW4 is out!

HW5 is a mini-project in groups of 2–4.
For HW4, submit your group members and proposed project ideas.

Check the course website for Office Hours:
https://harvard-cs-1840.github.io/2026/

Recap

Infinite horizon Discounted MDPs

$$\mathscr{M} = \{S, A, \gamma, r, P, \mu\}$$

Average state distribution: $d^{\pi} = (1-\gamma)\sum_{h=0}^{\infty} \gamma^h \mathbb{P}_h^{\pi}$

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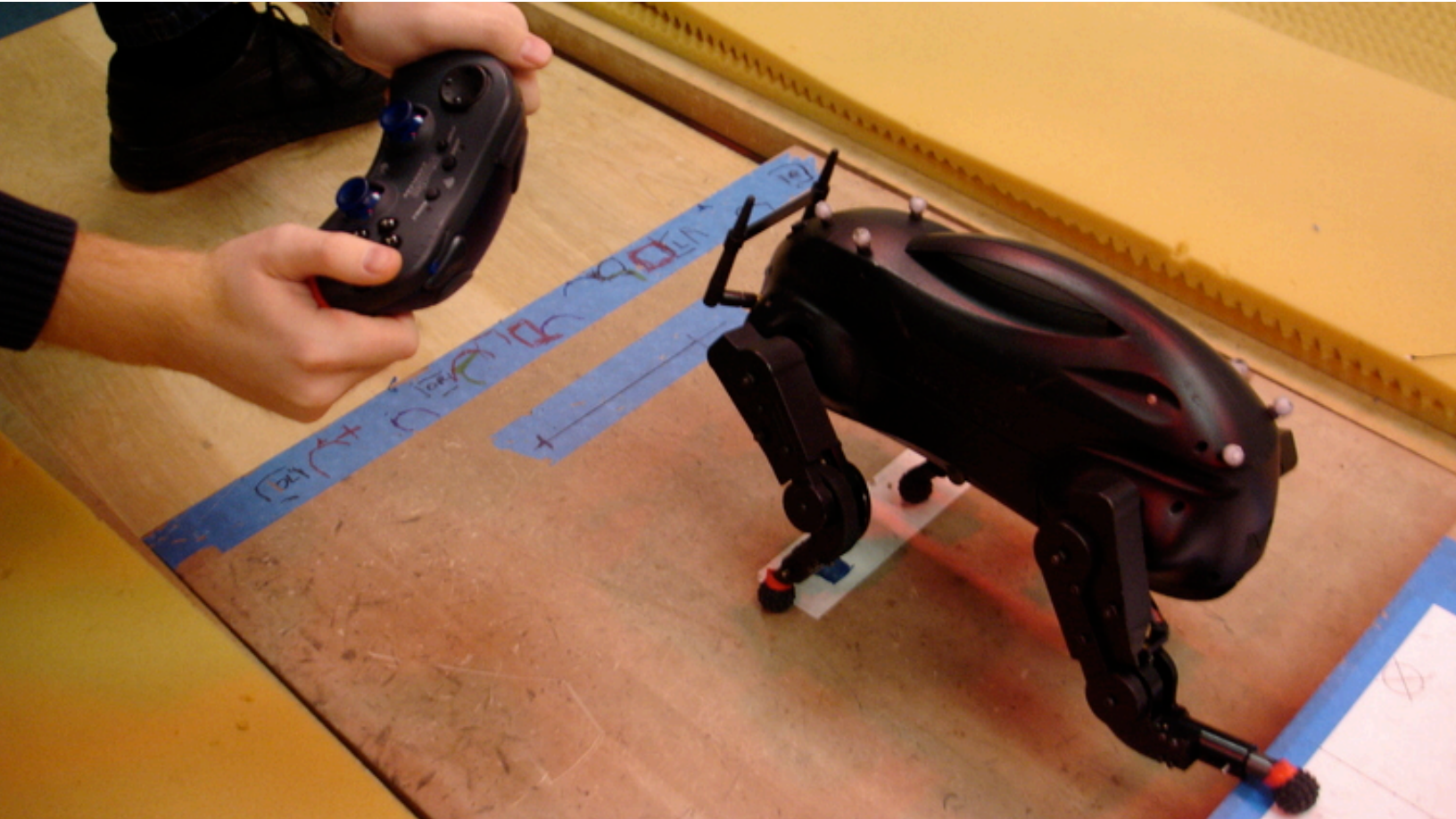
Average state distribution: $d^\pi = (1-\gamma)\sum_{h=0}^{\infty} \gamma^h \mathbb{P}_h^\pi$

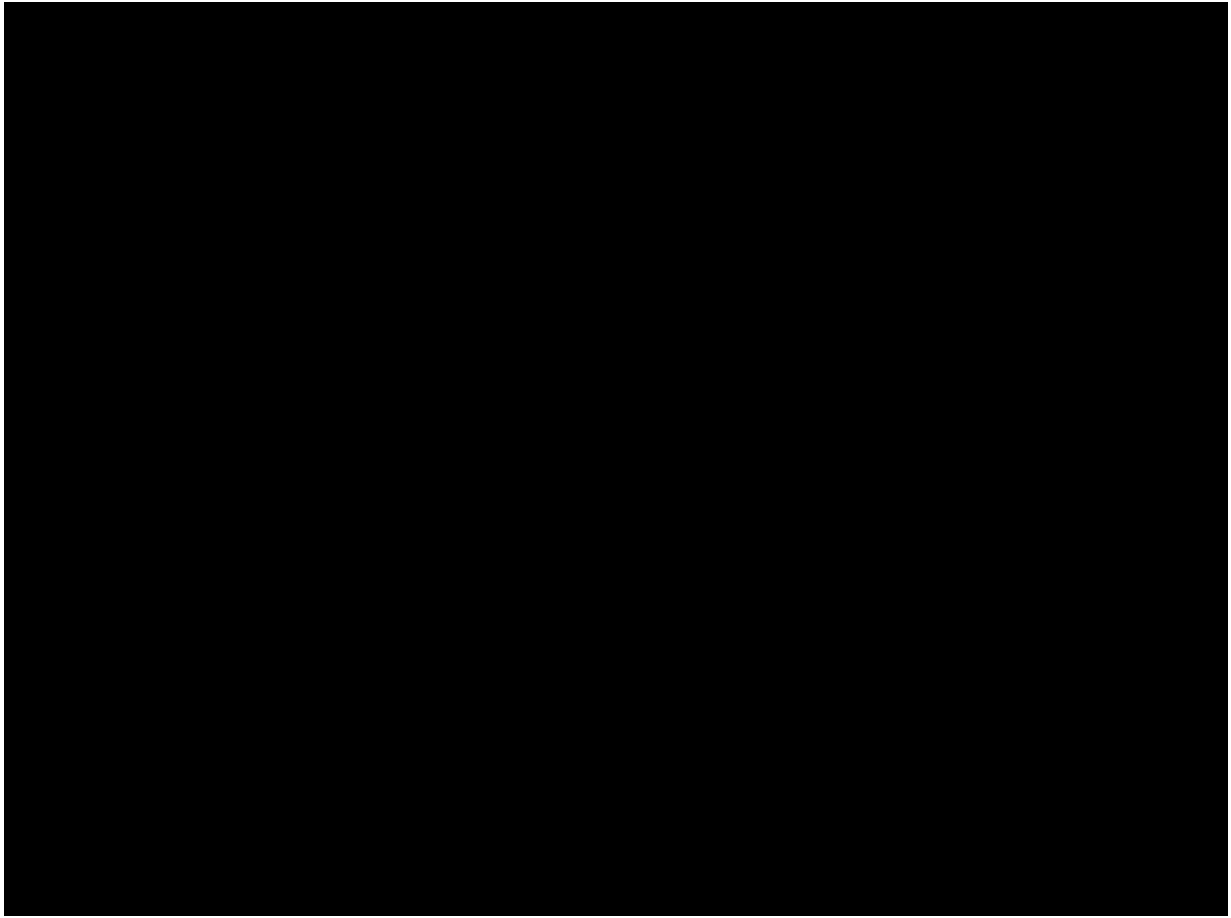
Advantage: $A^\pi := Q^\pi(s,a) - V^\pi(s,a)$

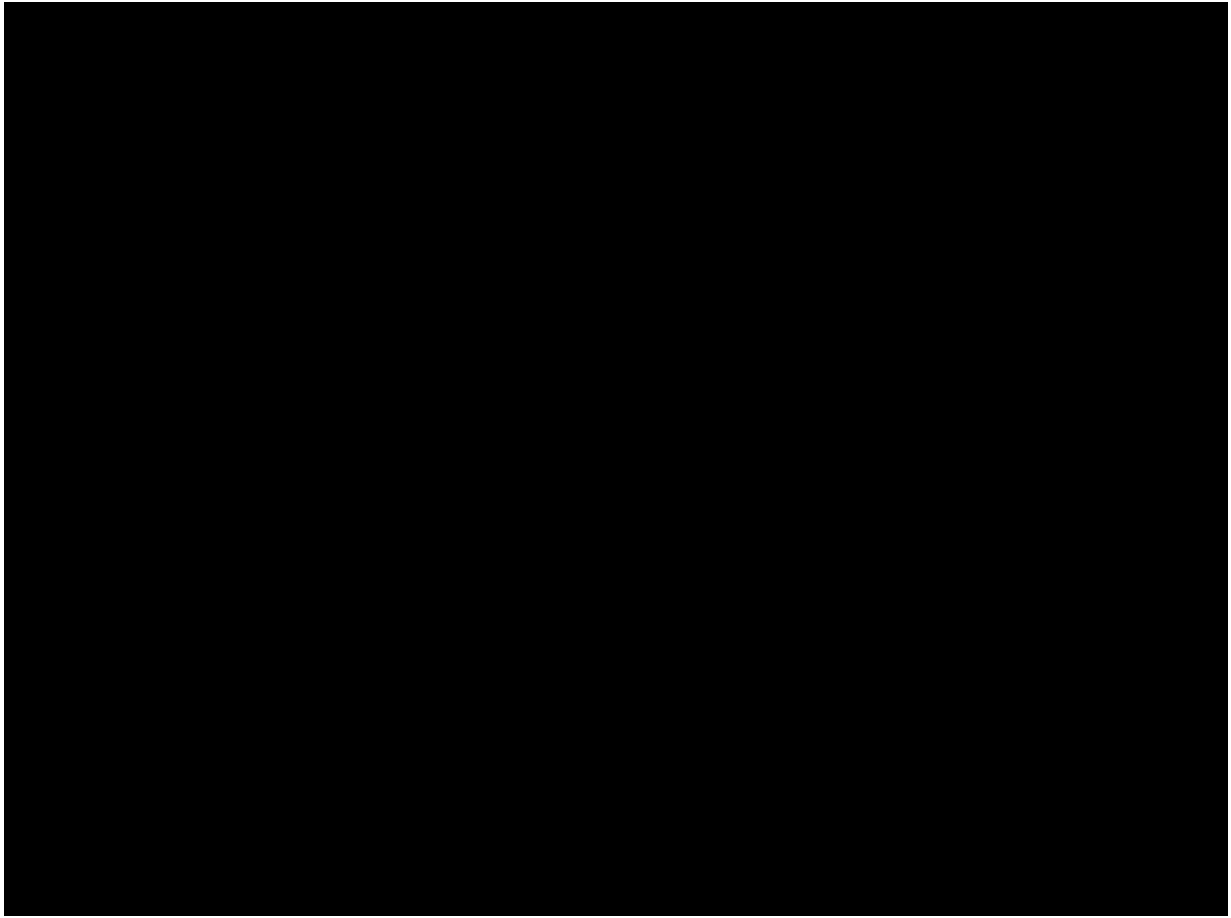
Outline:

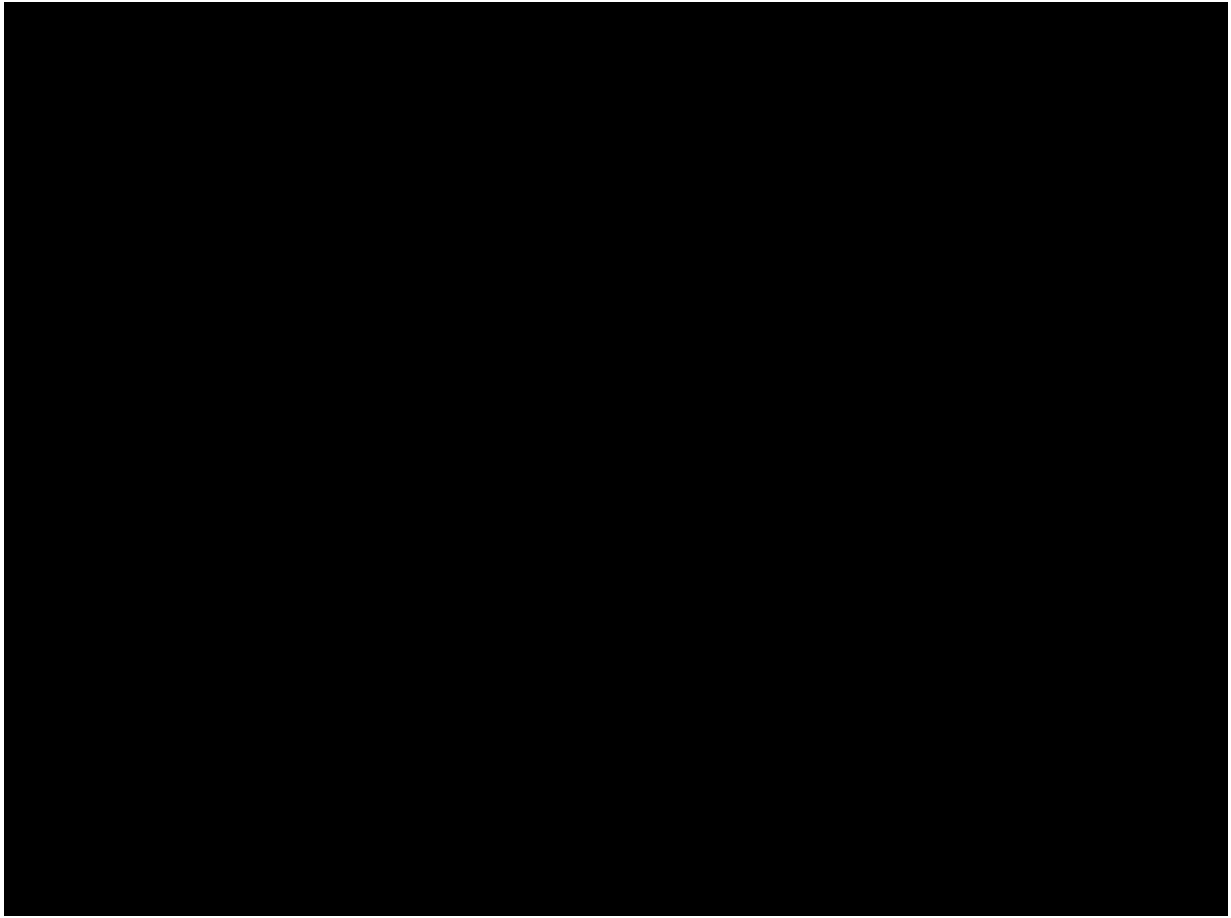
1. Offline Imitation Learning: Behavior Cloning

2. Performance difference lemma and its application to proving BC's bound









An Autonomous Land Vehicle

In A Neural Network [Pomerleau, NIPS '88]



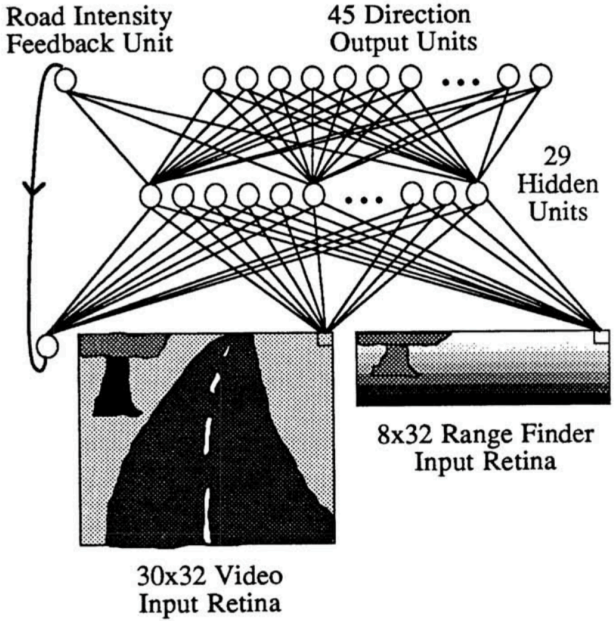


Figure 1: ALVINN Architecture

Imitation Learning

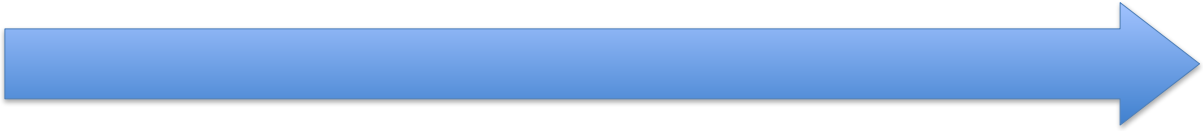


Imitation Learning

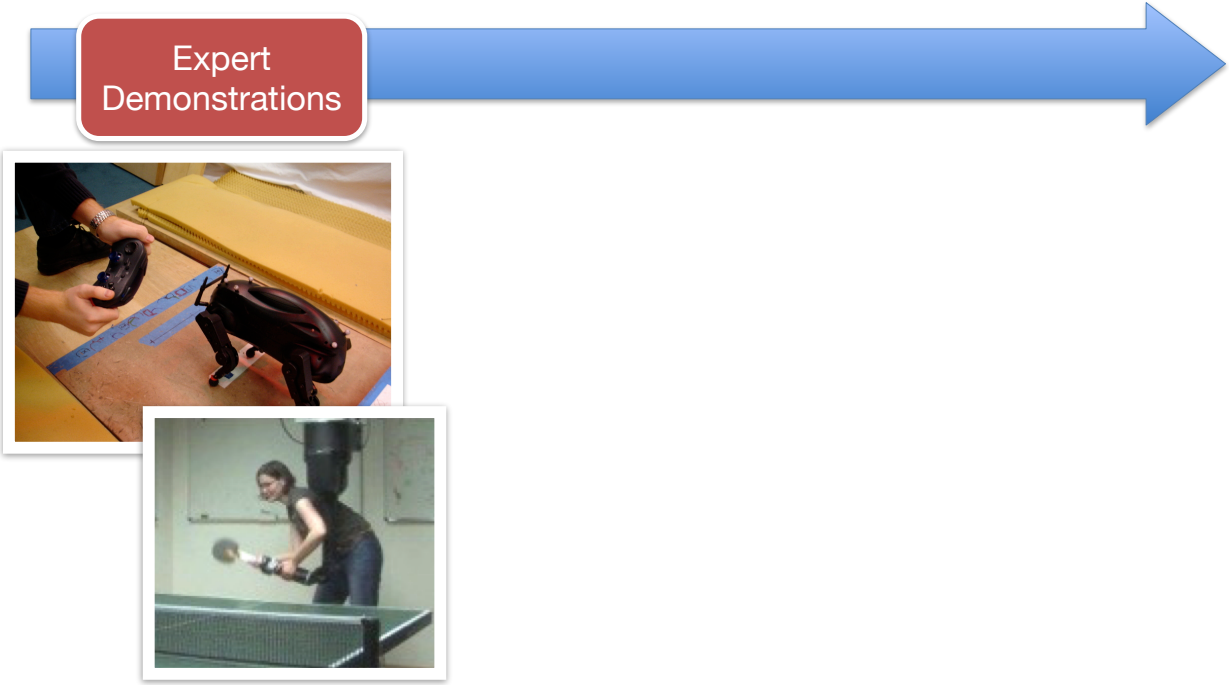


Imitation Learning

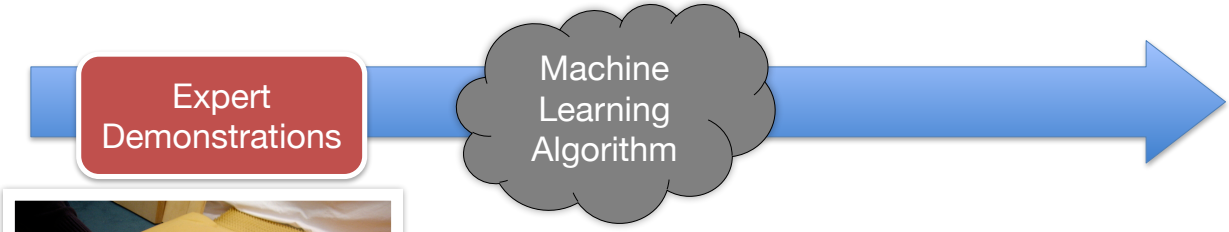
Imitation Learning



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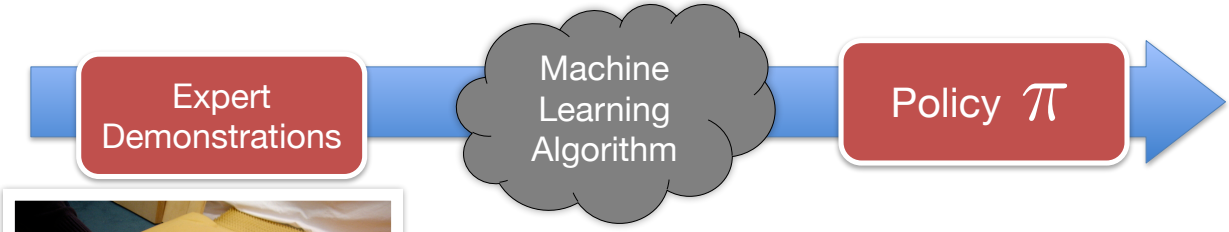
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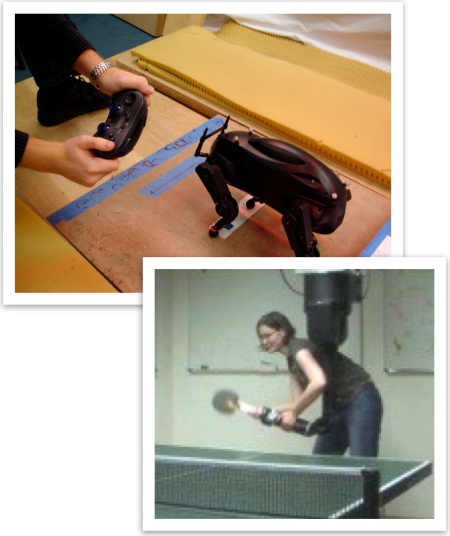




- SVM
- Gaussian Process
- Kernel Estimator
- Deep Networks
- Random Forests
- LWR
- …

Imitation Learning





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Maps *states* to <u>actions</u>

Learning to Drive by Imitation

[Pomerleau89, Saxena05, Ross11a]

Input:



Camera Image

Policy

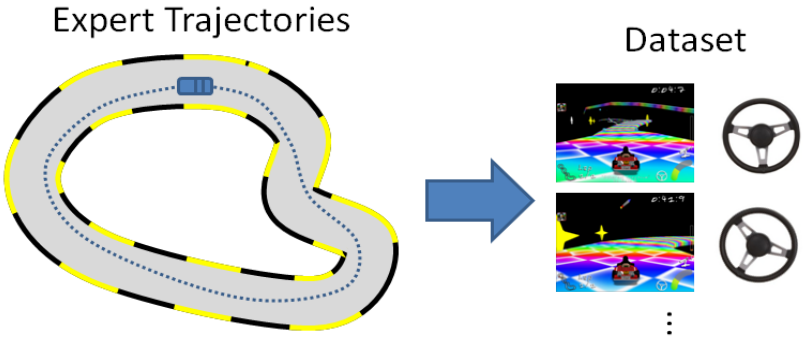
Output:



Steering Angle
in [-1, 1]

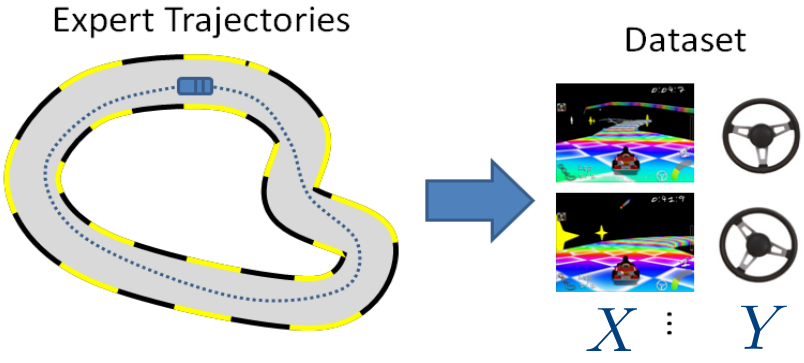
[Widrow64,Pomerleau89]

Supervised Learning Approach: Behavior Cloning



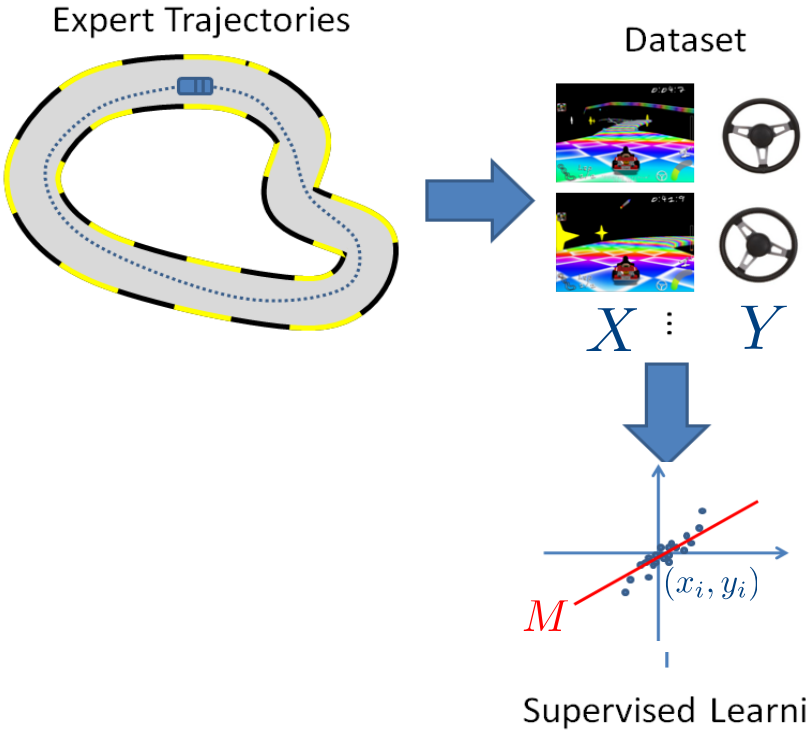
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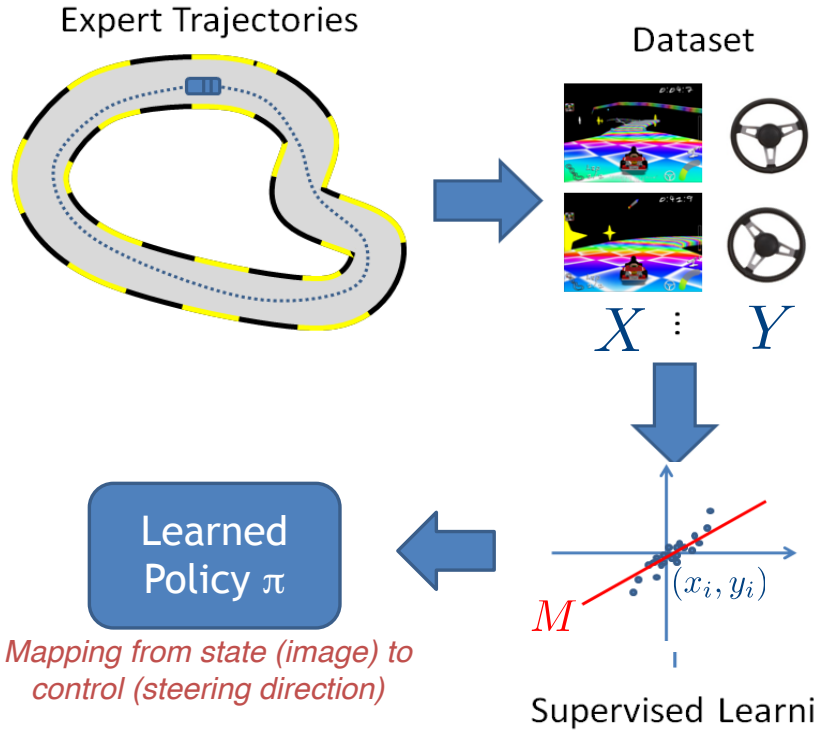
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Supervised Learning Approach: Behavior Cloning




Ready!


Ready!


Ready!

LLMs are trained via BC in their pre-training phase

Take a sentence from the web:

Reinforcement learning (RL) is an interdisciplinary area of machine learning and optimal control concerned with how an intelligent agent should take actions in a dynamic environment in order to maximize a reward signal.

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Reinforcement learning (RL) is an interdisciplinary area of machine learning and optimal control concerned with how an intelligent agent should take actions in a dynamic environment in order to maximize a reward signal.

Forcing LLM to predict the next "action" conditioned on past…

Let's formalize the offline IL Setting

and the Behavior Cloning algorithm

Discounted infinite horizon MDP $\mathscr{M} = \{S, A, \gamma, r, P, \rho, \pi^\star\}$

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We have a dataset $\mathscr{D} = (s_i^\star, a_i^\star)_{i=1}^M \sim d^{\pi^\star}$

$s^\star \sim d^{\pi^\star}$

$a^\star \sim \pi^\star(\cdot|s^\star)$

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We have a dataset $\mathscr{D} = (s_i^\star, a_i^\star)_{i=1}^M \sim d^{\pi^\star}$

Goal: learn a policy from $\mathscr{D}$ that is as good as the expert $\pi^\star$

Let's formalize the Behavior Cloning algorithm

BC Algorithm input: a restricted policy class $\Pi = \{\pi : S \mapsto \Delta(A)\}$

BC is a Reduction to Supervised Learning:

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BC is a Reduction to Supervised Learning:

$$\widehat{\pi} = \arg\min_{\pi \in \Pi} \sum_{i=1}^{M} \ell\left(\pi, s^\star, a^\star\right)$$

loss function

$D = \{(s_i^*, a_i^*)\}_{i=1}^{N}$

$s^* \sim d^{\pi^*}$

$a^* \sim \pi^*(\cdot | s^*)$

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Many choices of loss functions:

1. Negative log-likelihood (NLL): $\ell(\pi, s^\star, a^\star) = -\ln \pi(a^\star \mid s^\star)$

2. square loss (i.e., regression for continuous action): $\ell(\pi, s^\star, a^\star) = \|\pi(s^\star) - a^\star\|_2^2$

Analysis

$$\widehat{\pi} = \arg\min_{\pi \in \Pi} \sum_{i=1}^{M} \ell\left(\pi, s^{\star}, a^{\star}\right)$$

Assumption: we are going to assume Supervised Learning succeeded

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$$\mathbb{E}_{s \sim d^{\pi^\star}} \mathbf{1}\left[\hat{\pi}(s) \neq \pi^\star(s)\right] \leq \epsilon \in \mathbb{R}^+$$

↑ train distribution

- How does this ϵ correlate to the performance of $V^{\hat{\pi}}$?
- What is the performance difference between $V^{\hat{\pi}}$ and V^{π^*}?

Analysis

$$\hat{\pi} = \arg\min_{\pi \in \Pi} \sum_{i=1}^{M} \ell\left(\pi, s^\star, a^\star\right)$$

Assumption: we are going to assume Supervised Learning succeeded

Train $\quad \mathbb{E}_{s \sim d^{\pi^\star}} \mathbf{1}\left[\hat{\pi}(s) \neq \pi^\star(s)\right] \leq \epsilon \in \mathbb{R}^+$

Does that imply that $\hat{\pi}$ is a good policy? What's the performance difference between $\hat{\pi}$ and $\pi^\star$?

Test $\quad \mathbb{E}_{s \sim d^{\hat{\pi}}}\left[\sum_{h=0}^{H} r(s_h, a_h)\right]$

Outline:



1. Offline Imitation Learning: Behavior Cloning

2. Performance difference lemma and its application to proving BC's bound

Performance Difference Lemma

Given two policies $\pi : S \mapsto \Delta(A)$, $\pi' : S \mapsto \Delta(A)$, recall $V^{\pi}(s_0) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, \pi\right]$

Performance Difference Lemma

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Performance Difference Lemma (PDL):

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi}_{s_0}} \left[\mathbb{E}_{a \sim \pi(\cdot|s)} Q^{\pi'}(s,a) - V^{\pi'}(s) \right]$$

$$:= \frac{1}{1-\gamma} \mathbb{E}_{s \sim d^{\pi}_{s_0}} \left[\mathbb{E}_{a \sim \pi(\cdot|s)} A^{\pi'}(s,a) \right]$$

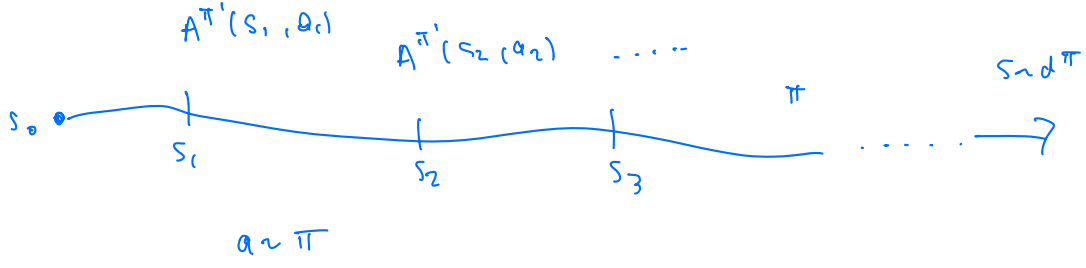
PDL Explanation

$$\mathbb{E}_{s\sim d^{\pi}, a\sim\pi}[A^{\pi}(s,a)] = 0$$

$$A^{\pi} = Q^{\pi} - V^{\pi}$$

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}A^{\pi'}(s,a)\right]$$

π π π'



PDL Proof

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}A^{\pi'}(s,a)\right]$$

Proof of Sketch (see reading material for detailed steps)

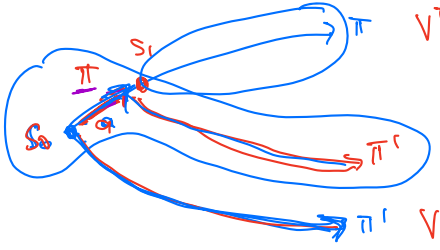
- Simulation lemma
- Monotonic Improvement

$V^{\pi}(s_0) - V^{\pi'}(s_0)$

$= ① - ③$

$= \boxed{① - ②} \boxed{+② - ③}$

$= V^{\pi}(s_1) - V^{\pi'}(s_1) + A^{\pi'}(s_0, a_0)$

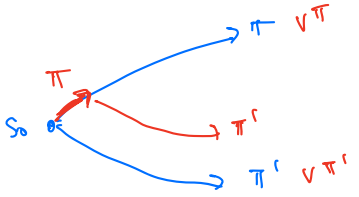


① $V^{\pi}(s_0)$

② $Q^{\pi'}(s, a)$ ↳ π

③ $V^{\pi'}(s_0)$

$A^{\pi'}(s,a) = Q^{\pi'}(s,a) - V^{\pi'}(s)$



PDL Proof

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}A^{\pi'}(s,a)\right]$$

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$$V^{\pi}(s_0) - V^{\pi'}(s_0)$$

$$= V^{\pi}(s_0) - \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[r(s_0,a_0) + \gamma\mathbb{E}_{s'\sim P(s_0,a_0)}V^{\pi'}(s')\right] + \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[r(s_0,a_0) + \gamma\mathbb{E}_{s'\sim P(s_0,a_0)}V^{\pi'}(s')\right] - V^{\pi'}(s_0)$$

(1) (2) (3)

BE

$= \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[r(s_0,a_0) + \gamma\mathbb{E}_{s'\sim P}\left[V^{\pi}(s')\right]\right]$

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↓ BE

$$= \gamma\mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\mathbb{E}_{s_1\sim P(s_0,a_0)}\left[V^{\pi}(s_1) - V^{\pi'}(s_1)\right] + \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[r(s_0,a_0) + \gamma\mathbb{E}_{s'\sim P(s_0,a_0)}V^{\pi'}(s')\right] - V^{\pi'}(s_0)$$

recursion

$Q^{\pi'}(s_0, a_0)$

$\mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[V^{\pi'}(s_0)\right]$

PDL Proof

$$V^{\pi}(s_0) - V^{\pi'}(s_0) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{a\sim\pi(\cdot|s)}A^{\pi'}(s,a)\right]$$

Proof of Sketch (see reading material for detailed steps)

$V^{\pi}(s_0) - V^{\pi'}(s_0)$ state s_0

$= V^{\pi}(s_0) - \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[r(s_0,a_0) + \gamma\mathbb{E}_{s'\sim P(s_0,a_0)}V^{\pi'}(s')\right] + \mathbb{E}_{a_0\sim\pi(\cdot|s_0)}\left[r(s_0,a_0) + \gamma\mathbb{E}_{s'\sim P(s_0,a_0)}V^{\pi'}(s')\right] - V^{\pi'}(s_0)$

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state s_1

$A^{\pi'}(s_0, a_0)$

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An Application of PDL in Policy Iteration

Recall that $\pi^{t+1}(s) = \arg\max_a A^{\pi^t}(s,a)$ $= \arg\max_a Q^{\pi_t}(s,a)$

Show monotonic improvement using PDL:

An Application of PDL in Policy Iteration

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PDL

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$a \sim \pi^{t+1}(\cdot|s)$

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$$\geq \frac{1}{1-\gamma}\mathbb{E}_{s\sim d_{s_0}^{\pi^{t+1}}}A^{\pi^t}(s,\pi^t(s))$$

π^{t+1} argmax $A^{\pi^t}(s,a)$

$A^{\pi^t}(s,\pi^t(s)) = 0$

$\mathbb{E}_{a\sim\pi}[A^{\pi}(s,a)] = 0$

An Application of PDL in Policy Iteration

Recall that $\pi^{t+1}(s) = \arg\max_a A^{\pi^t}(s,a)$

Show monotonic improvement using PDL:

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Analysis of BC

Theorem [BC Performance] With probability at least $1 - \delta$, BC returns a policy $\widehat{\pi}$:

$$V^{\pi^\star} - V^{\widehat{\pi}} \leq \frac{2}{(1-\gamma)^2}\epsilon$$

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The quadratic amplification is annoying;
Related to pre-trained LLM hallucination;
Will see how to fix it next lecture

Summary

BC: simple algorithm that directly learns from human demonstrations; used in robotics and NLP

PDL: how to capture the performance difference between two policies
(as important as Simulation lemma)

Feedback and Attendance

https://bit.ly/4gYn2nC

